

76

FOR IBPS SBI PO/CLERK PRELIMS 2026

QUANT CHECKLIST

the PRACTICE PAPER

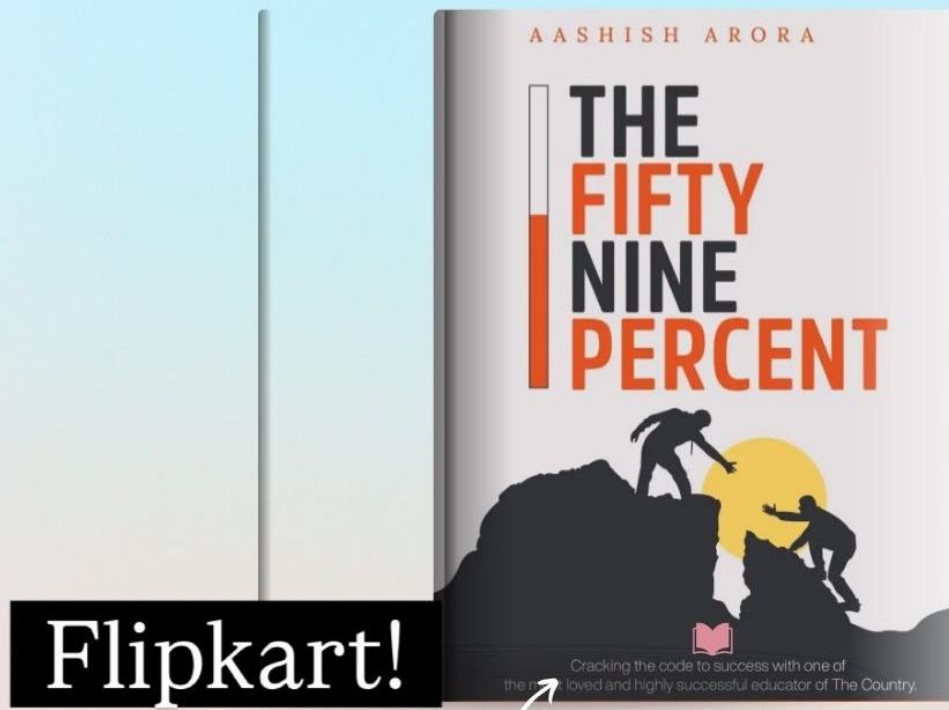


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Dear Students,

This daily practice sheet is designed to build both speed and accuracy, one day at a time.

It contains a mix of easy, moderate, and challenging questions to prepare you for every possible scenario in the exam. Treat it like a warm-up before the real game.

Solve it daily without fail. Don't wait for motivation—show up with discipline. Because it's not talent but consistent hard work that takes you places.

Stay focused. Stay consistent. Work hard.

-Aashish Arora

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1. SIMPLIFICATION AND APPROXIMATION

Direction: What value should come in place of the question mark (?) in the following question?

1. $58.33\% \text{ of } 1584 + 31.25\% \text{ of } 1024 - \sqrt{5184} = ? \times 14 + 52$

- A. 72
- B. 76
- C. 80
- D. 84
- E. None of these

2. $? \% \text{ of } 720 + \frac{4}{9} \text{ of } 405 = 75\% \text{ of } 960 - 18^2$

- A. 24
- B. 30
- C. 36
- D. 42
- E. None of these

3. $\frac{13}{16} \text{ of } 768 - \frac{7}{12} \text{ of } 600 + \sqrt[3]{21952} = ? + 130$

- A. 160
- B. 164

- C. 168
- D. 172
- E. None of these

4. $\sqrt{?} + 37.5\% \text{ of } 512 = 25^2 - 17^2$

- A. 20164
- B. 20736
- C. 21316
- D. 21904
- E. None of these

5. $4^4 + 5^3 + 6^2 = ? \times 11 + 10$

- A. 37
- B. 39
- C. 41
- D. 43
- E. None of these

6. $\sqrt[3]{125000} + \frac{7}{13} \text{ of } 949 - 22.22\% \text{ of } 540 = ? \div 5 + 120$

- A. 1485
- B. 1525
- C. 1565

D. 1605

E. None of these

$$7. 18.75 \times 64 - 6.25 \times 96 + 12.5 \times 48 = ? \times 24$$

A. 46

B. 50

C. 54

D. 58

E. None of these

$$8. 5\frac{3}{8} + 8\frac{1}{6} - 3\frac{5}{12} + 4\frac{1}{2} = ?$$

A. $14\frac{3}{8}$ B. $14\frac{5}{8}$ C. $15\frac{1}{8}$ D. $15\frac{3}{8}$

E. None of these

$$9. (\sqrt{7569} + \sqrt[3]{64000}) \times 8 - 29^2 = ? + 15$$

A. 160

B. 166

C. 172

D. 178

E. None of these

$$10. 30\% \text{ of } 40\% \text{ of } 1500 + 12.5\% \text{ of } 720 = ? \times 6$$

A. 39

B. 42

C. 45

D. 48

E. None of these

$$11. \frac{5}{11} \text{ of } 968 + \frac{9}{16} \text{ of } 1152 - \frac{7}{15} \text{ of } 645 = ? + 582$$

A. 190

B. 195

C. 200

D. 205

E. None of these

$$12. ? \times 14 - 83.33\% \text{ of } 432 = 24^2 + 20 \times 12$$

A. 84

B. 88

C. 92

D. 96

E. None of these

$$13. \sqrt[3]{117649} + 37.5\% \text{ of } 1344 + 15^2 = ? \times 8 + 18$$

A. 86

B. 90

C. 99

D. 104

E. None of these

14. $8160 \div 15 \div 8 + 9072 \div 12 \div 9 + \sqrt{784} = ?$

A. 168

B. 172

C. 176

D. 184

E. None of these

15. $\frac{2}{7}$ of 2450 - $\frac{5}{8}$ of 768 + 56.25% of 1152 = ? + 340

A. 520

B. 528

C. 536

D. 544

E. None of these

16. $25\% \text{ of } 960 + 6.25\% \text{ of } 5120 - \frac{1}{5} \text{ of } 750 = ? + 280$

A. 118

B. 122

C. 126

D. 130

E. None of these

17. $7\frac{5}{8} - 3\frac{3}{16} + 4\frac{11}{16} - 2\frac{1}{8} = ?$

A. 7

B. $7\frac{1}{2}$

C. 8

D. $8\frac{1}{2}$

E. None of these

18. $42 \times 18 + 56.25\% \text{ of } 320 - ?^2 = 15 \times 60$

A. 4

B. 5

C. 6

D. 7

E. None of these

19. $\{68.75\% \text{ of } 640 + \frac{13}{40} \text{ of } 800\} \div ? = 35$

A. 20

B. 22

C. 24

D. 26

E. None of these

20. $? \div 12 + 45.45\% \text{ of } 1320 = 75\% \text{ of } 960 + \sqrt[3]{592704}$

A. 2352

- B. 2400
C. 2424
D. 2448
E. None of these

Answers:

1. C
2. B
3. D
4. B
5. A
6. D
7. B
8. B
9. A
10. C
11. D
12. A
13. E
14. E
15. B
16. D
17. A
18. C
19. A
20. D

Solutions:

$$1. 58.33\% \text{ of } 1584 + 31.25\% \text{ of } 1024 - \sqrt{5184} = ? \times 14 + 52$$

$$58.33\% = \frac{7}{12}, 31.25\% = \frac{5}{16} \text{ and } \sqrt{5184} = 72$$

$$58.33\% \text{ of } 1584 = 924 \text{ and } 31.25\% \text{ of } 1024 = 320$$

$$\text{So, } 924 + 320 - 72 = ? \times 14 + 52$$

$$1172 = ? \times 14 + 52$$

$$? \times 14 = 1120$$

$$? = 80$$

$$2. ?\% \text{ of } 720 + \frac{4}{9} \text{ of } 405 = 75\% \text{ of } 960 - 18^2$$

$$\frac{4}{9} \text{ of } 405 = 180, 75\% \text{ of } 960 = 720 \text{ and } 18^2 = 324$$

$$\text{So, } ?\% \text{ of } 720 + 180 = 720 - 324$$

$$?\% \text{ of } 720 + 180 = 396$$

$$?\% \text{ of } 720 = 216$$

$$? = 216 \times 100 \div 720$$

$$? = 30$$

$$3. \frac{13}{16} \text{ of } 768 - \frac{7}{12} \text{ of } 600 + \sqrt[3]{21952} = ? + 130$$

$$\frac{13}{16} \text{ of } 768 = 624, \frac{7}{12} \text{ of } 600 = 350 \text{ and } \sqrt[3]{21952} = 28$$

$$\text{So, } 624 - 350 + 28 = ? + 130$$

$$302 = ? + 130$$

$$? = 172$$

$$4. \sqrt{?} + 37.5\% \text{ of } 512 = 25^2 - 17^2$$

$$37.5\% = \frac{3}{8}, 25^2 = 625 \text{ and } 17^2 = 289$$

$$37.5\% \text{ of } 512 = 192$$

$$\text{So, } \sqrt{?} + 192 = 625 - 289$$

$$\sqrt{?} + 192 = 336$$

$$\sqrt{?} = 144$$

$$? = 20736$$

$$5. 4^4 + 5^3 + 6^2 = ? \times 11 + 10$$

$$4^4 = 256, 5^3 = 125 \text{ and } 6^2 = 36$$

$$\text{So, } 256 + 125 + 36 = ? \times 11 + 10$$

$$417 = ? \times 11 + 10$$

$$? \times 11 = 407$$

$$? = 37$$

$$6. \sqrt[3]{125000} + \frac{7}{13} \text{ of } 949 - 22.22\% \text{ of } 540 = ? \div 5 + 120$$

$$\sqrt[3]{125000} = 50 \text{ and } 22.22\% = \frac{2}{9}$$

$$\frac{7}{13} \text{ of } 949 = 511 \text{ and } 22.22\% \text{ of } 540 = 120$$

$$\text{So, } 50 + 511 - 120 = ? \div 5 + 120$$

$$441 = ? \div 5 + 120$$

$$? \div 5 = 321$$

$$? = 1605$$

$$7. 18.75 \times 64 - 6.25 \times 96 + 12.5 \times 48 = ? \times 24$$

$$18.75 \times 64 = 1200, 6.25 \times 96 = 600 \text{ and } 12.5 \times 48 = 600$$

$$\text{So, } 1200 - 600 + 600 = ? \times 24$$

$$1200 = ? \times 24$$

$$? = 50$$

$$8. 5\frac{3}{8} + 8\frac{1}{6} - 3\frac{5}{12} + 4\frac{1}{2} = ?$$

$$\text{Whole number part} = 5 + 8 - 3 + 4 = 14$$

$$\text{Fraction part} = \frac{3}{8} + \frac{1}{6} - \frac{5}{12} + \frac{1}{2}$$

$$\text{After taking denominator 24, fraction part} = \frac{9}{24} + \frac{4}{24} - \frac{10}{24} + \frac{12}{24}$$

$$\text{Fraction part} = \frac{15}{24} = \frac{5}{8}$$

$$\text{So, } ? = 14 + \frac{5}{8}$$

$$? = 14\frac{5}{8}$$

$$9. (\sqrt{7569} + \sqrt[3]{64000}) \times 8 - 29^2 = ? + 15$$

$$\sqrt{7569} = 87, \sqrt[3]{64000} = 40 \text{ and } 29^2 = 841$$

$$\text{So, } (87 + 40) \times 8 - 841 = ? + 15$$

$$127 \times 8 - 841 = ? + 15$$

$$1016 - 841 = ? + 15$$

$$175 = ? + 15$$

$$? = 160$$

10. 30% of 40% of 1500 + 12.5% of 720 = ? × 6

30% of 40% of 1500 = 180 and 12.5% = $\frac{1}{8}$

12.5% of 720 = 90

So, 180 + 90 = ? × 6

270 = ? × 6

? = 45

11. $\frac{5}{11}$ of 968 + $\frac{9}{16}$ of 1152 - $\frac{7}{15}$ of 645 = ? + 582

$\frac{5}{11}$ of 968 = 440, $\frac{9}{16}$ of 1152 = 648 and $\frac{7}{15}$ of 645 = 301

So, 440 + 648 - 301 = ? + 582

787 = ? + 582

? = 205

12. ? × 14 - 83.33% of 432 = 24² + 20 × 12

83.33% = $\frac{5}{6}$, 24² = 576 and 20 × 12 = 240

83.33% of 432 = 360

So, ? × 14 - 360 = 576 + 240

? × 14 - 360 = 816

? × 14 = 1176

? = 84

13. $\sqrt[3]{117649} + 37.5\% \text{ of } 1344 + 15^2 = ? \times 8 + 18$

$\sqrt[3]{117649} = 49$, 37.5% = $\frac{3}{8}$ and 15² = 225

37.5% of 1344 = 504

So, 49 + 504 + 225 = ? × 8 + 18

778 = ? × 8 + 18

? × 8 = 760

? = 95

14. 8160 ÷ 15 ÷ 8 + 9072 ÷ 12 ÷ 9 + $\sqrt{784}$ = ?

8160 ÷ 15 ÷ 8 = 68, 9072 ÷ 12 ÷ 9 = 84 and $\sqrt{784} = 28$

So, 68 + 84 + 28 = ?

? = 180

15. $\frac{2}{7}$ of 2450 - $\frac{5}{8}$ of 768 + 56.25% of 1152 = ? + 340

$\frac{2}{7}$ of 2450 = 700, $\frac{5}{8}$ of 768 = 480 and 56.25% = $\frac{9}{16}$

56.25% of 1152 = 648

So, 700 - 480 + 648 = ? + 340

868 = ? + 340

? = 528

16. 25% of 960 + 6.25% of 5120 - $\frac{1}{5}$ of 750 = ? + 280

$$25\% = \frac{1}{4} \text{ and } 6.25\% = \frac{1}{16}$$

$$25\% \text{ of } 960 = 240, 6.25\% \text{ of } 5120 = 320 \text{ and } \frac{1}{5} \text{ of } 750 = 150$$

$$\text{So, } 240 + 320 - 150 = ? + 280$$

$$410 = ? + 280$$

$$? = 130$$

$$17. 7\frac{5}{8} - 3\frac{3}{16} + 4\frac{11}{16} - 2\frac{1}{8} = ?$$

$$\text{Whole number part} = 7 - 3 + 4 - 2 = 6$$

$$\text{Fraction part} = \frac{5}{8} - \frac{3}{16} + \frac{11}{16} - \frac{1}{8}$$

$$\text{After taking denominator 16, fraction part} = \frac{10}{16} - \frac{3}{16} + \frac{11}{16} - \frac{2}{16}$$

$$\text{Fraction part} = \frac{16}{16} = 1$$

$$\text{So, } ? = 6 + 1$$

$$? = 7$$

$$18. 42 \times 18 + 56.25\% \text{ of } 320 - ?^2 = 15 \times 60$$

$$42 \times 18 = 756, 56.25\% = \frac{9}{16} \text{ and } 15 \times 60 = 900$$

$$56.25\% \text{ of } 320 = 180$$

$$\text{So, } 756 + 180 - ?^2 = 900$$

$$936 - ?^2 = 900$$

$$?^2 = 36$$

$$? = 6$$

$$19. \{68.75\% \text{ of } 640 + \frac{13}{40} \text{ of } 800\} \div ? = 35$$

$$68.75\% = \frac{11}{16}$$

$$68.75\% \text{ of } 640 = 440 \text{ and } \frac{13}{40} \text{ of } 800 = 260$$

$$\text{So, } (440 + 260) \div ? = 35$$

$$700 \div ? = 35$$

$$? = 20$$

$$20. ? \div 12 + 45.45\% \text{ of } 1320 = 75\% \text{ of } 960 + \sqrt[3]{592704}$$

$$45.45\% = \frac{5}{11}, 75\% = \frac{3}{4} \text{ and } \sqrt[3]{592704} = 84$$

$$45.45\% \text{ of } 1320 = 600 \text{ and } 75\% \text{ of } 960 = 720$$

$$\text{So, } ? \div 12 + 600 = 720 + 84$$

$$? \div 12 + 600 = 804$$

$$? \div 12 = 204$$

$$? = 2448$$

2. ARITHMETIC QUESTIONS

Q1: Sonal started a business by investing Rs. 90000. After 2 months, Nisha joined her with Rs. 150000. After another 6 months, Sonal withdrew $\frac{1}{3}$ of her investment, and Nisha added Rs. 30000 more. At the end of the year, the total profit was Rs. 86000. Find Sonal's profit share.

सोनल ने Rs. 90000 निवेश करके एक व्यवसाय शुरू किया। 2 महीने बाद, निशा Rs. 150000 के साथ व्यवसाय में शामिल हुई। अगले 6 महीने बाद, सोनल ने अपने निवेश का $\frac{1}{3}$ भाग निकाल लिया और निशा ने Rs. 30000 और जोड़ दिए। वर्ष के अंत में कुल लाभ Rs. 86000 था। सोनल का लाभांश ज्ञात कीजिए।

- (A) Rs. 28000
- (B) Rs. 32000
- (C) Rs. 36000
- (D) Rs. 40000
- (E) Rs. 30000

Q2: Train P, running at 72 km/hr, crosses a platform whose length is half of its own length in 45 seconds. Find the time taken by Train P to cross Train Q, whose length is $\frac{5}{3}$ times the length of Train P and which is running at 54 km/hr in the same direction.

ट्रेन P, 72 किमी/घंटा की गति से चलते हुए, एक ऐसे प्लेटफॉर्म को 45 सेकंड में पार करती है जिसकी लंबाई ट्रेन P की अपनी लंबाई की आधी है। ट्रेन P को ट्रेन Q को पार करने में कितना समय लगेगा, जिसकी लंबाई ट्रेन P की लंबाई की $\frac{5}{3}$ गुनी है और जो उसी दिशा में 54 किमी/घंटा की गति से चल रही है?

- (A) 300 seconds
- (B) 280 seconds
- (C) 360 seconds
- (D) 320 seconds
- (E) 400 seconds

Q3: Arvind spends 35% of his monthly salary on rent and 25% on food. From the remaining amount, he gives money to his father and mother in the ratio of 7 : 5. If the difference between the amounts given to his father and mother is Rs. 10000, find Arvind's monthly salary.

अरविंद अपने मासिक वेतन का 35% किराए पर और 25% भोजन पर खर्च करता है। शेष राशि में से वह अपने पिता और माता को 7 : 5 के अनुपात में पैसे देता है। यदि पिता और माता को दी गई राशि का अंतर Rs. 10000 है, तो अरविंद का मासिक वेतन ज्ञात कीजिए।

- (A) Rs. 150000
- (B) Rs. 120000
- (C) Rs. 180000
- (D) Rs. 135000
- (E) Rs. 160000

Q4: The sum of the breadth of a rectangle and the side of a square is 39 meters. If the perimeter of the rectangle is 84 meters and the length of the rectangle is 26 meters, find the side of the square.

एक आयत की चौड़ाई और एक वर्ग की भुजा का योग 39 मीटर है। यदि आयत का परिमाप 84 मीटर है और आयत की लंबाई 26 मीटर है, तो वर्ग की भुजा ज्ञात कीजिए।

- (A) 19 meters
- (B) 21 meters
- (C) 25 meters
- (D) 18 meters
- (E) 23 meters

Q5: Pipe P can fill 20% of a tank in 4 hours, and Pipe Q can fill 25% of the tank in 5 hours. Pipe R is an outlet pipe and can empty the full tank in 40 hours. If the tank is already 25% filled, find the time taken by all three pipes together to fill the remaining part of the tank.

पाइप P एक टंकी का 20% भाग 4 घंटे में भर सकता है और पाइप Q टंकी का 25% भाग 5 घंटे में भर सकता है। पाइप R एक निकासी पाइप है और पूरी टंकी को 40 घंटे में खाली कर सकता है। यदि टंकी पहले से 25% भरी हुई है, तो तीनों पाइप मिलकर शेष भाग को भरने में कितना समय लेंगे?

- (A) 8 hours
- (B) 12 hours
- (C) 10 hours

(D) 15 hours

(E) 6 hours

Q6: A trader purchased a wristwatch for Rs P and sold it for Rs Q at a profit of 30%. If he had purchased it at 25% less and sold it for Rs 96 less, he would have earned a profit of 20%. Find the value of $Q - P$.

एक व्यापारी ने एक कलाई घड़ी Rs P में खरीदी और उसे Rs Q में 30% लाभ पर बेचा। यदि उसने उसे 25% कम मूल्य पर खरीदा होता और Rs 96 कम में बेचा होता, तो उसे 20% लाभ होता। $Q - P$ का मान ज्ञात कीजिए।

(A) 60

(B) 90

(C) 84

(D) 72

(E) 96

Q7: A vendor mixed $(x + 40)$ liters of water with 240 liters of milk. Then 60 liters of the mixture was replaced with pure milk. If the quantity of milk in the final mixture becomes 264 liters, find x .

एक विक्रेता ने 240 लीटर दूध में $(x + 40)$ लीटर पानी मिलाया। फिर मिश्रण में से 60 लीटर निकालकर उसकी जगह शुद्ध दूध मिला दिया गया। यदि अंतिम मिश्रण में दूध की मात्रा 264 लीटर हो जाती है, तो x ज्ञात कीजिए।

(A) 100

(B) 140

(C) 120

(D) 160

(E) 180

Q8: A boat covers 120 km downstream in t hours and 140 km upstream in $(t + 3)$ hours. If the ratio of the speed of the boat in still water to its downstream speed is 5:6, find the time taken by the boat to cover 60 km downstream and 60 km upstream.

एक नाव धारा के अनुकूल 120 किमी की दूरी t घंटों में और धारा के प्रतिकूल 140 किमी की दूरी $(t + 3)$ घंटों में तय करती है। यदि शांत जल में नाव की गति और धारा के अनुकूल गति का अनुपात 5:6 है, तो नाव द्वारा 60 किमी धारा के अनुकूल और 60 किमी धारा के प्रतिकूल

दूरी तय करने में लिया गया समय ज्ञात कीजिए।

- (A) 4 hours
- (B) 5 hours
- (C) 6 hours
- (D) 7 hours
- (E) 8 hours

Q9: Two buses start at the same time from two cities toward each other. The distance between the two cities is 360 km. The first bus can cover the whole distance in 6 hours, and the second bus can cover the same distance in 12 hours. After how many hours will they meet?

दो बसें दो शहरों से एक ही समय पर एक-दूसरे की ओर चलना शुरू करती हैं। दोनों शहरों के बीच की दूरी 360 किमी है। पहली बस पूरी दूरी 6 घंटों में तय कर सकती है और दूसरी बस वही दूरी 12 घंटों में तय कर सकती है। वे कितने घंटों बाद मिलेंगी?

- (A) 4 hours
- (B) 3 hours
- (C) 5 hours
- (D) 6 hours
- (E) 8 hours

Q10: A person invests Rs 20,000 in Scheme A at $r\%$ per annum simple interest for 3 years. The interest earned from Scheme A is invested in Scheme B at 10% per annum compound interest for 2 years. If the amount received from Scheme B is Rs 7,260, find the value of r .

एक व्यक्ति Rs 20,000 को योजना A में 3 वर्षों के लिए $r\%$ वार्षिक साधारण ब्याज पर निवेश करता है। योजना A से प्राप्त ब्याज को वह योजना B में 2 वर्षों के लिए 10% वार्षिक चक्रवृद्धि ब्याज पर निवेश करता है। यदि योजना B से प्राप्त राशि Rs 7,260 है, तो r का मान ज्ञात कीजिए।

- (A) 8
- (B) 12
- (C) 15
- (D) 9
- (E) 10

Q11: The average monthly income of Kartik and Lalit is Rs. 5600. The average monthly income of Lalit and Manav is Rs. 6200, and the average monthly income of Kartik and Manav is Rs. 5900. Find Kartik's monthly income.

कार्तिक और ललित की औसत मासिक आय Rs. 5600 है। ललित और मानव की औसत मासिक आय Rs. 6200 है तथा कार्तिक और मानव की औसत मासिक आय Rs. 5900 है। कार्तिक की मासिक आय ज्ञात कीजिए।

- (A) Rs. 5100
- (B) Rs. 5400
- (C) Rs. 5200
- (D) Rs. 5300
- (E) Rs. 5500

Q12: Nitin had a certain amount. He invested 60% of it in a mutual fund, 15% in a fixed deposit, and 10% in gold. If the remaining amount with him is Rs. 7500, find the total amount he had initially.

नितिन के पास एक निश्चित राशि थी। उसने उसका 60% म्यूचुअल फंड में, 15% फिक्स्ड डिपॉजिट में और 10% सोने में निवेश किया। यदि उसके पास शेष राशि Rs. 7500 है, तो प्रारंभ में उसके पास कुल कितनी राशि थी?

- (A) Rs. 45000
- (B) Rs. 50000
- (C) Rs. 60000
- (D) Rs. 40000
- (E) Rs. 55000

Q13: Nirav started a business with Rs. 60000. After 3 months, Pallav joined him with Rs. 72000. After 5 more months, Uday joined them with Rs. 90000. If the total profit at the end of one year is Rs. 144000, find Uday's profit share.

निरव ने Rs. 60000 से एक व्यवसाय शुरू किया। 3 महीने बाद पल्लव Rs. 72000 के साथ उससे जुड़ गया। इसके 5 महीने बाद उदय Rs. 90000 के साथ उनसे जुड़ गया। यदि एक वर्ष के अंत में कुल लाभ Rs. 144000 है, तो उदय का लाभांश ज्ञात कीजिए।

- (A) Rs. 30000
- (B) Rs. 27000
- (C) Rs. 36000
- (D) Rs. 24000

(E) Rs. 32000

Q14: A trader marks his goods 50% above the cost price and allows a discount of 20%. He cheats by giving only 800 grams instead of 1 kg. Find his gain percent.

एक व्यापारी अपने सामान का अंकित मूल्य क्रय मूल्य से 50% अधिक रखता है और 20% की छूट देता है। वह 1 किग्रा के स्थान पर केवल 800 ग्राम सामान देकर धोखा देता है। उसका लाभ प्रतिशत ज्ञात कीजिए।

- (A) 40%
- (B) 45%
- (C) 50%
- (D) 55%
- (E) 60%

Q15: Ketan, Lokesh and Nilesh together can complete a work in 15 days, and Lokesh alone can complete the same work in 40 days. Find in how many days Ketan and Nilesh together can complete the whole work.

केतन, लोकेश और निलेश मिलकर किसी कार्य को 15 दिनों में पूरा कर सकते हैं, और लोकेश अकेला उसी कार्य को 40 दिनों में पूरा कर सकता है। केतन और निलेश मिलकर पूरा कार्य कितने दिनों में पूरा कर सकते हैं?

- (A) 20 days
- (B) 18 days
- (C) 30 days
- (D) 25 days
- (E) 24 days

Q16: A boat travels 216 km upstream in 18 hours and 300 km downstream in 20 hours. Find the time taken by the boat in still water to cover 81 km.

एक नाव 18 घंटों में धारा के प्रतिकूल 216 किमी और 20 घंटों में धारा के अनुकूल 300 किमी चलती है। स्थिर जल में 81 किमी की दूरी तय करने में नाव को कितना समय लगेगा?

- (A) 4 hours
- (B) 5 hours
- (C) 6 hours
- (D) 8 hours

(E) 9 hours

Q17: A mixture contains 120 liters of milk and water in the ratio 2 : 3, respectively. If 25% of the mixture is taken out and X liters of milk are added to the remaining mixture, then the ratio of milk to water becomes 4 : 3. Find X.

एक मिश्रण में 120 लीटर दूध और पानी क्रमशः 2 : 3 के अनुपात में हैं। यदि मिश्रण का 25% निकाल लिया जाता है और बचे हुए मिश्रण में x लीटर दूध मिलाया जाता है, तो दूध और पानी का अनुपात 4 : 3 हो जाता है। x ज्ञात कीजिए।

- (A) 24
- (B) 30
- (C) 36
- (D) 42
- (E) 48

Q18: The area of a square is 784 sq. cm, and its side is equal to the side of a cube. Find the total surface area of the cube.

एक वर्ग का क्षेत्रफल 784 वर्ग सेमी है और उसकी भुजा एक घन की भुजा के बराबर है। घन का कुल पृष्ठीय क्षेत्रफल ज्ञात कीजिए।

- (A) 4200
- (B) 4480
- (C) 4704
- (D) 4900
- (E) 5040

Q19: The number of students in School N is 25% more than that in School O, and the number of students in School M is 30% less than that in School O. If the total number of students in Schools M and O together is 204, find the total number of students in Schools N and O together.

स्कूल N में छात्रों की संख्या स्कूल O से 25% अधिक है, और स्कूल M में छात्रों की संख्या स्कूल O से 30% कम है। यदि स्कूल M और O में कुल छात्रों की संख्या 204 है, तो स्कूल N और O में कुल छात्रों की संख्या ज्ञात कीजिए।

- (A) 240
- (B) 260
- (C) 270

(D) 280

(E) 300

Q20: Partner U and Partner V started a business by investing Rs. $3x$ and Rs. $5x$, respectively. After 4 months, Partner W joined the business with an investment of Rs. $6x$. If the total profit at the end of the year is Rs. 66000, find the share of Partner V in the profit.

पार्टनर U और पार्टनर V ने क्रमशः Rs. $3x$ और Rs. $5x$ का निवेश करके एक व्यवसाय शुरू किया। 4 महीने बाद, पार्टनर W ने Rs. $6x$ के निवेश के साथ व्यवसाय में प्रवेश किया। यदि वर्ष के अंत में कुल लाभ Rs. 66000 है, तो पार्टनर V का लाभ हिस्सा ज्ञात कीजिए।

(A) Rs. 24000

(B) Rs. 27500

(C) Rs. 30000

(D) Rs. 32500

(E) Rs. 25000

Answers:

1. B	11.D
2. D	12.B
3. A	13.A
4. E	14.C
5. C	15.E
6. D	16.C
7. C	17.C
8. B	18.C
9. A	19.C
10.E	20.B

Solutions:

1. Option B

In partnership, profit is divided in the ratio of capital $\times$ time.

Sonal invested Rs. 90000 for the first 8 months.

Sonal's capital-time value for 8 months = $90000 \times 8 = 720000$

After 8 months, Sonal withdrew $\frac{1}{3}$ of her investment.

Amount withdrawn = $90000 \times \frac{1}{3} = \text{Rs. } 30000$

Remaining investment of Sonal = $90000 - 30000 = \text{Rs. } 60000$

Sonal's capital-time value for the last 4 months = $60000 \times 4 = 240000$

Total capital-time value of Sonal = $720000 + 240000 = 960000$

Nisha joined after 2 months with Rs. 150000.

Nisha invested Rs. 150000 for the next 6 months.

Nisha's capital-time value for 6 months = $150000 \times 6 = 900000$

After that, Nisha added Rs. 30000.

New investment of Nisha = $150000 + 30000 = \text{Rs. } 180000$

Nisha's capital-time value for the last 4 months = $180000 \times 4 = 720000$

Total capital-time value of Nisha = $900000 + 720000 = 1620000$

Profit ratio of Sonal and Nisha = $960000 : 1620000$

Divide both terms by 60000.

Profit ratio = $16 : 27$

Total ratio units = $16 + 27 = 43$

Total profit = Rs. 86000

Value of 1 unit = $86000/43 = \text{Rs. } 2000$

Sonal's profit share = $16 \times 2000 = \text{Rs. } 32000$

2. Option D

Speed of Train P = 72 km/hr

Convert km/hr into m/sec.

Speed of Train P = $72 \times \frac{5}{18} = 20 \text{ m/sec}$

Let the length of Train P be L meters.

Length of platform = $L/2$

Distance covered while crossing the platform = $L + L/2 = 3L/2$

Train P crosses the platform in 45 seconds.

So, distance = Speed $\times$ Time

$3L/2 = 20 \times 45$

$3L/2 = 900$

$L = 900 \times \frac{2}{3} = 600 \text{ meters}$

Length of Train P = 600 meters

Length of Train Q = $\frac{5}{3}$ of Train P

Length of Train Q = $600 \times \frac{5}{3} = 1000 \text{ meters}$

Speed of Train Q = 54 km/hr

Speed of Train Q = $54 \times \frac{5}{18} = 15$ m/sec

Both trains are moving in the same direction.

Relative speed = $20 - 15 = 5$ m/sec

Total distance to cross each other = $600 + 1000 = 1600$ meters

Time taken = Distance/Relative speed

Time taken = $1600/5 = 320$ seconds

3. Option A

Arvind spends 35% on rent and 25% on food.

Total spending = $35\% + 25\% = 60\%$

Remaining amount = $100\% - 60\% = 40\%$ of salary

The remaining amount is divided between father and mother in the ratio of 7 : 5.

Difference in ratio parts = $7 - 5 = 2$ parts

Given difference = Rs. 10000

So, 2 parts = Rs. 10000

1 part = $10000/2 =$ Rs. 5000

Total ratio parts = $7 + 5 = 12$ parts

Remaining amount = $12 \times 5000 =$ Rs. 60000

This remaining amount is 40% of salary.

So, 40% of salary = Rs. 60000

Salary = $60000 \times 100/40 =$ Rs. 150000

4. Option E

Perimeter of rectangle = 84 meters

Formula of perimeter = $2 \times (\text{Length} + \text{Breadth})$

So, $2 \times (\text{Length} + \text{Breadth}) = 84$

Length + Breadth = $84/2 = 42$ meters

Length of rectangle = 26 meters

Breadth of rectangle = $42 - 26 = 16$ meters

Given, breadth of rectangle + side of square = 39 meters

Side of square = $39 - 16 = 23$ meters

5. Option C

Pipe P fills 20% of the tank in 4 hours.

$$20\% = 20/100 = 1/5$$

Work done by Pipe P in 1 hour = $1/5 \times 1/4 = 1/20$

Pipe Q fills 25% of the tank in 5 hours.

$$25\% = 25/100 = 1/4$$

Work done by Pipe Q in 1 hour = $1/4 \times 1/5 = 1/20$

Pipe R empties the full tank in 40 hours.

Work done by Pipe R in 1 hour = $-1/40$

Net work done by all three pipes in 1 hour = $1/20 + 1/20 - 1/40$

LCM of 20, 20 and 40 = 40

$$\text{Net work} = 2/40 + 2/40 - 1/40 = 3/40$$

Tank is already 25% filled.

$$\text{Remaining part} = 75\% = 75/100 = 3/4$$

Required time = Remaining work/Net one-hour work

$$\text{Required time} = 3/4 \times 40/3 = 10 \text{ hours}$$

6. Option D

Let the cost price of the wristwatch be Rs P.

Selling price = Rs Q.

Profit = 30%.

$$\text{So, } Q = 130/100 \times P$$

$$Q = 13P/10$$

In the second case, the cost price is 25% less.

$$\text{New cost price} = 75/100 \times P$$

$$\text{New cost price} = 3P/4$$

Profit in the second case = 20%.

$$\text{New selling price} = 120/100 \times 3P/4$$

$$\text{New selling price} = 6/5 \times 3P/4$$

$$\text{New selling price} = 18P/20$$

$$\text{New selling price} = 9P/10$$

Given, new selling price is Rs 96 less than Q.

$$\text{So, } Q - 96 = 9P/10$$

$$\text{Put } Q = 13P/10.$$

$$13P/10 - 96 = 9P/10$$

$$13P/10 - 9P/10 = 96$$

$$4P/10 = 96$$

$$2P/5 = 96$$

$$P = 96 \times 5/2$$

$$P = 240$$

Now, $Q - P$ is the profit.

Profit = 30% of P

$$Q - P = 30/100 \times 240$$

$$Q - P = 72$$

7. Option C

Quantity of milk = 240 liters.

Quantity of water added = $(x + 40)$ liters.

Total mixture = $240 + x + 40 = (x + 280)$ liters.

60 liters of mixture is removed.

Milk removed = $60 \times 240/(x + 280)$

After this, 60 liters of pure milk is added.

Final milk = Initial milk – Milk removed + Milk added

$$\text{Final milk} = 240 - 60 \times 240/(x + 280) + 60$$

Given final milk = 264 liters.

$$\text{So, } 240 + 60 - 60 \times 240/(x + 280) = 264$$

$$300 - 60 \times 240/(x + 280) = 264$$

$$60 \times 240/(x + 280) = 36$$

$$14400/(x + 280) = 36$$

$$x + 280 = 14400/36$$

$$x + 280 = 400$$

$$x = 400 - 280$$

$$x = 120$$

8. Option B

Let the speed of the boat in still water be $5a$.

Downstream speed = $6a$.

So, speed of stream = $6a - 5a = a$.

Upstream speed = $5a - a = 4a$.

Downstream speed : Upstream speed = $6a : 4a = 3 : 2$

Downstream distance = 120 km and time = t hours.

Upstream distance = 140 km and time = $(t + 3)$ hours.

So, downstream speed/upstream speed = $3/2$.

$$(120/t)/(140/(t+3)) = 3/2$$

$$120 \times (t+3)/(140t) = 3/2$$

Cross multiply:

$$240(t+3) = 420t$$

$$240t + 720 = 420t$$

$$720 = 180t$$

$$t = 720/180$$

$$t = 4$$

$$\text{Downstream speed} = 120/4 = 30 \text{ km/h.}$$

$$\text{Upstream time} = t + 3 = 4 + 3 = 7 \text{ hours.}$$

$$\text{Upstream speed} = 140/7 = 20 \text{ km/h.}$$

$$\text{Time taken to cover 60 km downstream} = 60/30 = 2 \text{ hours.}$$

$$\text{Time taken to cover 60 km upstream} = 60/20 = 3 \text{ hours.}$$

$$\text{Total time} = 2 + 3 = 5 \text{ hours.}$$

9. Option A

Distance between the two cities = 360 km.

The first bus covers 360 km in 6 hours.

$$\text{Speed of first bus} = 360/6 = 60 \text{ km/h.}$$

The second bus covers 360 km in 12 hours.

$$\text{Speed of second bus} = 360/12 = 30 \text{ km/h.}$$

They move toward each other.

$$\text{Relative speed} = 60 + 30 = 90 \text{ km/h.}$$

$$\text{Time taken to meet} = \text{Distance/Relative speed}$$

$$\text{Time taken} = 360/90$$

$$\text{Time taken} = 4 \text{ hours.}$$

10. Option E

Amount received from Scheme B = Rs 7,260.

Scheme B gives compound interest at 10% per annum for 2 years.

Let the amount invested in Scheme B be Rs I.

$$\text{Amount after 2 years} = I \times 110/100 \times 110/100$$

$$7260 = I \times 11/10 \times 11/10$$

$$7260 = I \times 121/100$$

$$I = 7260 \times 100/121$$

$$I = 6000$$

So, interest earned from Scheme A = Rs 6000.

In Scheme A:

Principal = Rs 20,000

Time = 3 years

Rate = r%

Simple Interest = Principal $\times$ Rate $\times$ Time/100

$$6000 = 20000 \times r \times 3/100$$

$$6000 = 600r$$

$$r = 6000/600$$

$$r = 10$$

11. Option D

Average income of Kartik and Lalit = Rs. 5600

So, Kartik + Lalit = 5600×2 = Rs. 11200

Average income of Lalit and Manav = Rs. 6200

So, Lalit + Manav = 6200×2 = Rs. 12400

Average income of Kartik and Manav = Rs. 5900

So, Kartik + Manav = 5900×2 = Rs. 11800

Now:

$$2 \times \text{Kartik} = (\text{Kartik} + \text{Lalit}) + (\text{Kartik} + \text{Manav}) - (\text{Lalit} + \text{Manav})$$

$$2 \times \text{Kartik} = 11200 + 11800 - 12400$$

$$2 \times \text{Kartik} = 10600$$

$$\text{Kartik} = 10600/2$$

$$\text{Kartik} = \text{Rs. } 5300$$

12. Option B

Let the total amount be 100%.

Amount invested in mutual fund = 60%

Amount invested in fixed deposit = 15%

Amount invested in gold = 10%

Total invested amount = $60\% + 15\% + 10\% = 85\%$

Remaining amount = $100\% - 85\% = 15\%$

Given remaining amount = Rs. 7500

So, 15% of total amount = Rs. 7500

Total amount = $7500 \times 100/15$

Total amount = Rs. 50000

13. Option A

Profit is divided in the ratio of capital $\times$ time.

Nirav invested Rs. 60000 for 12 months.

Nirav's capital-months = $60000 \times 12 = 720000$

Pallav joined after 3 months, so he invested for 9 months.

Pallav's capital-months = $72000 \times 9 = 648000$

Uday joined after $3 + 5 = 8$ months, so he invested for 4 months.

Uday's capital-months = $90000 \times 4 = 360000$

Profit ratio = $720000 : 648000 : 360000$

Profit ratio = $10 : 9 : 5$

Total ratio parts = $10 + 9 + 5 = 24$

Total profit = Rs. 144000

Uday's share = $144000 \times 5/24$

Uday's share = Rs. 30000

14. Option C

Let the cost price of 1 kg goods be Rs. 100.

Marked price = 50% above cost price

Marked price = Rs. 150

Discount = 20%

Selling price charged for 1 kg = 80% of 150

Selling price = $80 \times 150/100 = \text{Rs. } 120$

But the trader gives only 800 grams.

Cost price of 800 grams = $800 \times 100/1000$

Cost price of 800 grams = Rs. 80

Profit = $120 - 80 = \text{Rs. } 40$

Gain percent = $40 \times 100/80$

Gain percent = 50%

15. Option E

Ketan, Lokesh and Nilesh together can complete the work in 15 days.

Their 1-day work = $1/15$

Lokesh alone can complete the work in 40 days.

Lokesh's 1-day work = $1/40$

Ketan and Nilesh's 1-day work = $1/15 - 1/40$

LCM of 15 and 40 = 120

Ketan and Nilesh's 1-day work = $8/120 - 3/120$

Ketan and Nilesh's 1-day work = $5/120$

Ketan and Nilesh's 1-day work = $1/24$

Therefore, Ketan and Nilesh together can complete the whole work in 24 days.

16. Option C

Upstream speed = $216/18 = 12$ km/hr.

Downstream speed = $300/20 = 15$ km/hr.

Speed of boat in still water = $(\text{Upstream speed} + \text{Downstream speed})/2$.

Speed of boat in still water = $(12 + 15)/2$.

Speed of boat in still water = $27/2$ km/hr.

Distance to be covered in still water = 81 km.

Time taken = Distance/Speed.

Time taken = $81/(27/2)$.

Time taken = $81 \times 2/27$.

Time taken = 6 hours.

17. Option C

Total mixture = 120 liters.

Ratio of milk to water = 2 : 3.

Total ratio parts = $2 + 3 = 5$.

Initial milk = $120 \times 2/5 = 48$ liters.

Initial water = $120 \times 3/5 = 72$ liters.

25% of the mixture is taken out.

Quantity taken out = $120 \times 25/100 = 30$ liters.

Milk removed = $30 \times 2/5 = 12$ liters.

Water removed = $30 \times 3/5 = 18$ liters.

Remaining milk = $48 - 12 = 36$ liters.

Remaining water = $72 - 18 = 54$ liters.

Now X liters of milk are added.

New milk = $36 + X$.

Water remains = 54 liters.

New ratio of milk to water = 4 : 3.

So,

$$(36 + X)/54 = 4/3$$

$$3(36 + X) = 54 \times 4$$

$$108 + 3X = 216$$

$$3X = 216 - 108$$

$$3X = 108$$

$$X = 108/3$$

$$X = 36$$

18. Option C

Area of square = 784 sq. cm.

Side of square = square root of 784 = 28 cm.

Side of cube = side of square = 28 cm.

Total surface area of cube = $6 \times \text{side} \times \text{side}$.

Total surface area = $6 \times 28 \times 28$.

Total surface area = 6×784 .

Total surface area = 4704 sq. cm.

19. Option C

Let the number of students in School O be x.

Students in School N = 25% more than School O.

Students in School N = 125% of x = $125x/100 = 5x/4$.

Students in School M = 30% less than School O.

Students in School M = 70% of x = $70x/100 = 7x/10$.

Given:

School M + School O = 204

$$7x/10 + x = 204$$

$$17x/10 = 204$$

$$x = 204 \times 10/17$$

$$x = 120$$

Students in School O = 120.

Students in School N = $5x/4 = 5 \times 120/4 = 150$.

Students in Schools N and O together = $150 + 120 = 270$.

20. Option B

Profit is divided in the ratio of investment $\times$ time.

Partner U invested Rs. $3x$ for 12 months.

Partner U's investment value = $3x \times 12 = 36x$.

Partner V invested Rs. $5x$ for 12 months.

Partner V's investment value = $5x \times 12 = 60x$.

Partner W joined after 4 months.

So, Partner W invested for 8 months.

Partner W's investment value = $6x \times 8 = 48x$.

Profit ratio of U, V and W = $36x : 60x : 48x$.

Simplified ratio = $3 : 5 : 4$.

Total ratio parts = $3 + 5 + 4 = 12$.

Total profit = Rs. 66000.

Partner V's share = $5/12 \times 66000$.

Partner V's share = Rs. 27500.

3. Quadratic Equations

In each of the following questions, there are two equations. You have to solve both equations and mark the correct answer.

(a) $x > y$

(b) $x < y$

(c) $x = y$ or the relationship cannot be established

(d) $x \geq y$

(e) $x \leq y$

1.) I. $10x^2 - 43x - 66 = 0$

II. $9y^2 + 51y + 60 = 0$

2.) I. $x^2 - 2\sqrt{2}x + 2 = 0$

II. $y^2 - 4\sqrt{2}y + 6 = 0$

3.) I. $12x^2 - 19x - 18 = 0$

II. $4y^2 - 31y + 55 = 0$

4.) I. $4x^2 + 5x - 21 = 0$

II. $5y^2 + 26y + 24 = 0$

5.) I. $x^2 - 4\sqrt{3}x + 12 = 0$

II. $2y^2 - 4\sqrt{3}y + 6 = 0$

6.) I. $8x^2 - 54x + 70 = 0$

II. $12y^2 + 3y - 42 = 0$

7.) I. $x^2 - 3\sqrt{2}x + 4 = 0$

II. $2y^2 - 12\sqrt{2}y + 36 = 0$

8.) I. $5x^2 - 17x - 12 = 0$

II. $7y^2 - 63y + 140 = 0$

9.) I. $3x^2 - 22x + 35 = 0$

II. $2y^2 + 9y + 4 = 0$

10.) I. $x^2 - 4\sqrt{3}x + 9 = 0$

II. $y^2 - 3\sqrt{3}y + 6 = 0$

11.) I. $6x^2 + 10x - 4 = 0$

II. $8y^2 - 44y + 48 = 0$

12.) I. $x^2 - 5\sqrt{2}x + 12 = 0$

II. $y^2 - 4\sqrt{2}y + 8 = 0$

13.) I. $7x^2 - 39x - 18 = 0$

II. $10y^2 - 55y + 25 = 0$

14.) I. $2x^2 - 17x + 35 = 0$

II. $11y^2 - 22y - 33 = 0$

15.) I. $x^2 - 2\sqrt{5}x + 5 = 0$

II. $y^2 - 4\sqrt{5}y + 15 = 0$

16.) I. $9x^2 - 54x + 72 = 0$

II. $3y^2 - 3y - 6 = 0$

17.) I. $11x^2 + 7x - 4 = 0$

II. $13y^2 - 39y + 26 = 0$

18.) I. $x^2 - 4\sqrt{7}x + 28 = 0$

II. $y^2 - 2\sqrt{7}y + 7 = 0$

19.) I. $12x^2 - 16x - 28 = 0$

II. $6y^2 - 38y + 56 = 0$

20.) I. $x^2 - 5\sqrt{3}x + 12 = 0$

II. $y^2 - 5\sqrt{3}y + 18 = 0$

Answers:

1. A

2. E

3. B

4. C

5. A

6. D

7. B

8. E

9. A

10. C

11. B

12. D

13. C

14. A

15. E

16. D

17. B

18. A

19. E

20. C

Solutions:

$$(1) x = -\frac{6}{5}, \frac{11}{2}$$

$$y = -4, -\frac{5}{3}$$

$$(2) x = \sqrt{2}, \sqrt{2}$$

$$y = \sqrt{2}, 3\sqrt{2}$$

$$(3) x = -\frac{2}{3}, \frac{9}{4}$$

$$y = \frac{11}{4}, 5$$

$$(4) x = -3, \frac{7}{4}$$

$$y = -4, -\frac{6}{5}$$

$$(5) x = 2\sqrt{3}, 2\sqrt{3}$$

$$y = \sqrt{3}, \sqrt{3}$$

$$(6) x = \frac{7}{4}, 5$$

$$y = -2, \frac{7}{4}$$

$$(7) x = \sqrt{2}, 2\sqrt{2}$$

$$y = 3\sqrt{2}, 3\sqrt{2}$$

$$(8) x = -\frac{3}{5}, 4$$

$$y = 4, 5$$

$$(9) x = \frac{7}{3}, 5$$

$$y = -4, -\frac{1}{2}$$

$$(10) x = \sqrt{3}, 3\sqrt{3}$$

$$y = \sqrt{3}, 2\sqrt{3}$$

$$(11) x = -2, \frac{1}{3}$$

$$y = \frac{3}{2}, 4$$

$$(12) x = 2\sqrt{2}, 3\sqrt{2}$$

$$y = 2\sqrt{2}, 2\sqrt{2}$$

$$(13) x = -\frac{3}{7}, 6$$

$$y = \frac{1}{2}, 5$$

$$(14) x = \frac{7}{2}, 5$$

$$y = -1, 3$$

$$(15) x = \sqrt{5}, \sqrt{5}$$

$$y = \sqrt{5}, 3\sqrt{5}$$

$$(16) x = 2, 4$$

$$y = -1, 2$$

$$(17) x = -1, \frac{4}{11}$$

$$y = 1, 2$$

$$(18) x = 2\sqrt{7}, 2\sqrt{7}$$

$$y = \sqrt{7}, \sqrt{7}$$

$$(19) x = -1, \frac{7}{3}$$

$$y = \frac{7}{3}, 4$$

$$(20) x = \sqrt{3}, 4\sqrt{3}$$

$$y = 2\sqrt{3}, 3\sqrt{3}$$

4. WRONG NUMBER SERIES

(1) 1540, 1558, 1585, 1628, 1674, 1740

- (a) 1558
- (b) 1628
- (c) 1674
- (d) 1740
- (e) None of these

(2) 2400, 2381, 2358, 2308, 2250, 2175

- (a) 2381
- (b) 2308
- (c) 2358
- (d) 2175
- (e) None of these

(3) 112, 255, 537, 1097, 2221, 4441

- (a) 537
- (b) 1097
- (c) 255
- (d) 2221
- (e) None of these

(4) 124, 362, 1053, 3131, 9358, 28032

- (a) 362
- (b) 3131
- (c) 1053
- (d) 28032
- (e) None of these

(5) 4000, 2040, 1060, 565, 325, 202.5

- (a) 2040
- (b) 1060
- (c) 325
- (d) 202.5
- (e) None of these

(6) 24822, 8250, 2745, 909, 300, 100

- (a) 909
- (b) 8250
- (c) 300
- (d) 100
- (e) None of these

(7) 670, 688, 721, 780, 848, 950

- (a) 688
- (b) 848

- (c) 780
- (d) 950
- (e) None of these

(8) 2600, 2564, 2504, 2414, 2298, 2120

- (a) 2564
- (b) 2414
- (c) 2120
- (d) 2298
- (e) None of these

(9) 104, 421, 1689, 6791, 27184, 108761

- (a) 1689
- (b) 421
- (c) 6791
- (d) 27184
- (e) None of these

(10) 79032, 19770, 3979, 686, 128, 52

- (a) 19770
- (b) 686
- (c) 128
- (d) 52
- (e) None of these

(11) 900, 920, 936, 970, 1020, 1090

- (a) 936
- (b) 920
- (c) 1020
- (d) 1090
- (e) None of these

(12) 550, 563, 595, 672, 794, 1015

- (a) 563
- (b) 794
- (c) 672
- (d) 1015
- (e) None of these

(13) 395, 431, 469, 509, 556, 595

- (a) 431
- (b) 469
- (c) 595
- (d) 556
- (e) None of these

(14) 2500, 2380, 2282, 2185, 2110, 2050

- (a) 2282
- (b) 2380
- (c) 2110
- (d) 2050
- (e) None of these

(15) 653, 727, 805, 895, 973, 1063

(a) 727

(b) 805

(c) 973

(d) 1063

(e) None of these

(16) 210, 228, 460, 480, 960, 990

(a) 228

(b) 460

(c) 960

(d) 990

(e) None of these

(17) 680.25, 696.75, 719.5, 753.5,
783.75, 825.25

(a) 696.75

(b) 783.75

(c) 753.5

(d) 825.25

(e) None of these

(18) 128, 640, 160, 800, 210, 1000

(a) 640

(b) 160

(c) 1000

(d) 210

(e) None of these

(19) 700, 725, 756, 824, 956, 1216

(a) 725

(b) 756

(c) 956

(d) 1216

(e) None of these

(20) 130, 251, 735, 2920, 14529,
87129

(a) 251

(b) 735

(c) 14529

(d) 87129

(e) None of these

Answers:

(1) b

(2) c

(3) d

(4) a

(5) e

(6) b

(7) c

(8) d

(9) a

(10) e

(11) b

(12) c

(13) d

(14) a

(15) e

(16) b

(17) c

(18) d

(19) a

(20) e

Solutions:

(1) $+(4^2 + 2)$, $+(5^2 + 2)$, $+(6^2 + 2)$, $+(7^2 + 2)$, $+(8^2 + 2)$.

(2) $-(4^2 + 3)$, $-(5^2 + 5)$, $-(6^2 + 7)$, $-(7^2 + 9)$, $-(8^2 + 11)$.

(3) $\times 2 + 31$, $\times 2 + 27$, $\times 2 + 23$, $\times 2 + 19$, $\times 2 + 15$.

(4) $\times 3 - 14$, $\times 3 - 21$, $\times 3 - 28$, $\times 3 - 35$, $\times 3 - 42$.

(5) $\div 2 + 40$, $\div 2 + 40$, $\div 2 + 40$, $\div 2 + 40$, $\div 2 + 40$.

(6) $\div 3 - 12$, $\div 3 - 9$, $\div 3 - 6$, $\div 3 - 3$, $\div 3$.

(7) $+(2 \times 9)$, $+(3 \times 11)$, $+(4 \times 13)$, $+(5 \times 15)$, $+(6 \times 17)$.

(8) $-(3 \times 12)$, $-(4 \times 15)$, $-(5 \times 18)$, $-(6 \times 21)$, $-(7 \times 24)$.

(9) $\times 4 + 5$, $\times 4 + 10$, $\times 4 + 15$, $\times 4 + 20$, $\times 4 + 25$.

(10) $\div 4 + 12$, $\div 5 + 18$, $\div 6 + 24$, $\div 7 + 30$, $\div 8 + 36$.

(11) $+14$, $+22$, $+34$, $+50$, $+70$; second differences: $+8$, $+12$, $+16$, $+20$.

(12) $+(2^3 + 5)$, $+(3^3 + 5)$, $+(4^3 + 5)$, $+(5^3 + 5)$, $+(6^3 + 5)$.

(13) $+36$, $+38$, $+40$, $+42$, $+44$.

(14) -120 , -105 , -90 , -75 , -60 .

(15) $18^2 \times 2 + 5$, $19^2 \times 2 + 5$, $20^2 \times 2 + 5$, $21^2 \times 2 + 5$, $22^2 \times 2 + 5$, $23^2 \times 2 + 5$.

(16) $+18$, $\times 2$, $+24$, $\times 2$, $+30$.

(17) $+16.5$, $+22.75$, $+29$, $+35.25$, $+41.5$.

(18) $\times 5$, $\div 4$, $\times 5$, $\div 4$, $\times 5$.

(19) $+20$, $+36$, $+68$, $+132$, $+260$; each next difference = previous difference $\times 2 - 4$.

(20) $\times 2 - 9$, $\times 3 - 18$, $\times 4 - 27$, $\times 5 - 36$, $\times 6 - 45$.

5. MISSING NUMBER SERIES

(1) 4680, 4667, 4642, ?, 4540, 4455

- (a) 4597
- (b) 4601
- (c) 4605
- (d) 4609
- (e) None of these

(2) 5, 14, 47, ?, 977, 5870

- (a) 188
- (b) 190
- (c) 192
- (d) 194
- (e) None of these

(3) 63, 88, 137, ?, 339, 508

- (a) 218
- (b) 220
- (c) 222
- (d) 224
- (e) None of these

(4) 8602, 4308, 1444, ?, 84, 25

- (a) 368
- (b) 370
- (c) 372
- (d) 374
- (e) None of these

(5) 24, 41, 73, ?, 215, 361

- (a) 120
- (b) 122
- (c) 124

(d) 126

(e) None of these

(6) 118, 124.5, 137.5, ?, 183, 215.5

- (a) 155
- (b) 157
- (c) 159
- (d) 161

(e) None of these

(7) 735, 721, 693, ?, 525, 301

- (a) 633
- (b) 635
- (c) 637
- (d) 639
- (e) None of these

(8) 18, 39, 87, ?, 483, 1209

- (a) 201
- (b) 203
- (c) 205
- (d) 207
- (e) None of these

(9) 302, 314, 342, ?, 462, 562

- (a) 384
- (b) 386
- (c) 388
- (d) 390
- (e) None of these

(10) 9, 24, 84, ?, 1794, 10794

- (a) 352
- (b) 354
- (c) 356
- (d) 358
- (e) None of these

(11) 410, 435, 475, ?, 600, 685

- (a) 530
- (b) 532
- (c) 534
- (d) 536
- (e) None of these

(12) 9362, 4683, 1564, ?, 84, 20

- (a) 389
- (b) 391
- (c) 393
- (d) 395
- (e) None of these

(13) 48, 68, 98, ?, 188, 248

- (a) 136
- (b) 138
- (c) 140
- (d) 142
- (e) None of these

(14) 128, 146, 178, ?, 300, 398

- (a) 228
- (b) 230
- (c) 232
- (d) 234
- (e) None of these

(15) 21, 43, 89, ?, 541, 1523

- (a) 203
- (b) 205
- (c) 207
- (d) 209
- (e) None of these

(16) 1525, 1505, 1475, ?, 1377, 1305

- (a) 1427
- (b) 1429
- (c) 1431
- (d) 1433
- (e) None of these

(17) 250, 265, 310, ?, 850, 2065

- (a) 443
- (b) 445
- (c) 447
- (d) 449
- (e) None of these

(18) 41, 85, 155, ?, 397, 581

- (a) 257
- (b) 259
- (c) 261
- (d) 263
- (e) None of these

(19) 3900, 3894, 3870, ?, 3690, 3480

- (a) 3806
- (b) 3808
- (c) 3810
- (d) 3812
- (e) None of these

(20) 32, 68, 152, ?, 744, 1588

- (a) 334
- (b) 336
- (c) 338
- (d) 340
- (e) None of these

Answers:

- (1) b
- (2) d

- (3) a
 (4) b
 (5) d
 (6) b
 (7) c
 (8) a
 (9) d
 (10) b
 (11) a
 (12) d
 (13) b
 (14) a
 (15) c
 (16) d
 (17) b
 (18) a
 (19) c
 (20) d

Solutions:

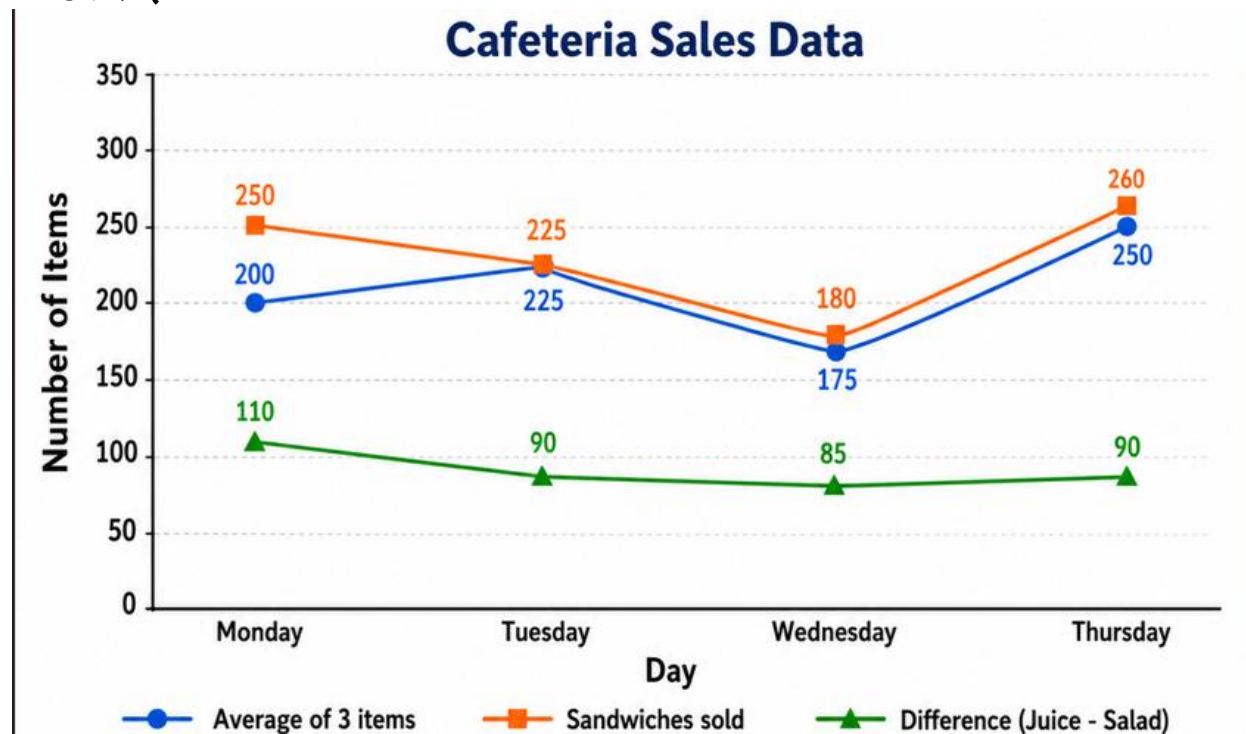
- (1) $-(3^2 + 2^2)$, $-(4^2 + 3^2)$, $-(5^2 + 4^2)$,
 $-(6^2 + 5^2)$, $-(7^2 + 6^2)$
 (2) $\times 2 + 4$, $\times 3 + 5$, $\times 4 + 6$, $\times 5 + 7$, $\times 6 + 8$
 (3) $+5^2$, $+7^2$, $+9^2$, $+11^2$, $+13^2$
 (4) $\div 2 + 7$, $\div 3 + 8$, $\div 4 + 9$, $\div 5 + 10$, $\div 6 + 11$
 (5) Sum of previous two numbers $+8$,
 $+12$, $+16$, $+20$
 (6) $+6.5$, $+13$, $+19.5$, $+26$, $+32.5$

- (7) -14 , -28 , -56 , -112 , -224
 (8) $\times 2 + 3$, $\times 2 + 9$, $\times 2 + 27$, $\times 2 + 81$, $\times 2 + 243$
 (9) $+12$, $+28$, $+48$, $+72$, $+100$
 Second difference: $+16$, $+20$, $+24$, $+28$
 (10) $\times 2 + 6$, $\times 3 + 12$, $\times 4 + 18$, $\times 5 + 24$,
 $\times 6 + 30$
 (11) $+25$, $+40$, $+55$, $+70$, $+85$
 (12) $\div 2 + 2$, $\div 3 + 3$, $\div 4 + 4$, $\div 5 + 5$, $\div 6 + 6$
 (13) $+20$, $+30$, $+40$, $+50$, $+60$
 (14) $+18$, $+32$, $+50$, $+72$, $+98$
 Second difference: $+14$, $+18$, $+22$, $+26$
 (15) $\times 3 - 20$, $\times 3 - 40$, $\times 3 - 60$, $\times 3 - 80$,
 $\times 3 - 100$
 (16) $-(4^2 + 4)$, $-(5^2 + 5)$, $-(6^2 + 6)$, $-(7^2 + 7)$, $-(8^2 + 8)$
 (17) $+15$, $+45$, $+135$, $+405$, $+1215$
 (18) Terms are: $n^3 + n^2 + 5$, for $n = 3$,
 4 , 5 , 6 , 7 , 8
 (19) $-1 \times 2 \times 3$, $-2 \times 3 \times 4$, $-3 \times 4 \times 5$,
 $-4 \times 5 \times 6$, $-5 \times 6 \times 7$
 (20) $\times 2 + 2^2$, $\times 2 + 4^2$, $\times 2 + 6^2$, $\times 2 + 8^2$,
 $\times 2 + 10^2$

6. DATA INTERPRETATION

Set 1: The line graph shows the average number of three food items, i.e., sandwiches, juice bottles and salad bowls sold by a cafeteria on four different days. It also shows the number of sandwiches sold and the difference between juice bottles and salad bowls sold by the cafeteria. On each day, the number of juice bottles sold is more than the number of salad bowls sold. Study the line graph carefully and answer the following questions.

रेखा आलेख चार अलग-अलग दिनों में एक कैफेटेरिया द्वारा बेचे गए तीन खाद्य पदार्थों, अर्थात् सैंडविच, जूस बोतलें और सलाद बाउल की औसत संख्या को दर्शाता है। यह सैंडविच की संख्या और जूस बोतलों तथा सलाद बाउल के बीच अंतर भी दर्शाता है। प्रत्येक दिन बेची गई जूस बोतलों की संख्या, बेचे गए सलाद बाउल की संख्या से अधिक है। रेखा आलेख का ध्यानपूर्वक अध्ययन करें और निम्नलिखित प्रश्नों के उत्तर दें।



Q1: The total number of food items sold on Saturday is 20% less than the total number of food items sold on Monday, and the number of salad bowls sold on Saturday is the same as that on Monday. If the ratio of sandwiches to juice bottles

sold on Saturday is 7 : 5, then find the number of sandwiches sold on Saturday.

शनिवार को बेचे गए कुल खाद्य पदार्थों की संख्या, सोमवार को बेचे गए कुल खाद्य पदार्थों की संख्या से 20% कम है, और शनिवार को बेचे गए सलाद बाउल की संख्या सोमवार के समान है। यदि शनिवार को बेचे गए सैंडविच और जूस बोतलों का अनुपात 7 : 5 है, तो शनिवार को बेचे गए सैंडविच की संख्या ज्ञात कीजिए।

- (A) 196
- (B) 224
- (C) 240
- (D) 210
- (E) 252

Q2: If the selling price of each sandwich, juice bottle and salad bowl sold on Wednesday is Rs. 12, Rs. 8 and Rs. A respectively, and the total revenue generated on Wednesday is Rs. 5700, then find the value of A.

यदि बुधवार को बेचे गए प्रत्येक सैंडविच, जूस बोतल और सलाद बाउल का विक्रय मूल्य क्रमशः Rs. 12, Rs. 8 और Rs. A है, और बुधवार को कुल आय Rs. 5700 है, तो A का मान ज्ञात कीजिए।

- (A) 12
- (B) 14
- (C) 16
- (D) 18
- (E) 20

Q3: Find the number of juice bottles sold on Tuesday as a percentage of the total number of sandwiches and salad bowls sold on the same day.

मंगलवार को बेची गई जूस बोतलों की संख्या, उसी दिन बेचे गए सैंडविच और सलाद बाउल की कुल संख्या का कितना प्रतिशत है?

- (A) 75%
- (B) $62 \frac{1}{2}\%$
- (C) $66 \frac{2}{3}\%$
- (D) 80%
- (E) 72%

Q4: If the average number of salad bowls sold on Monday, Tuesday, Wednesday, Thursday and Friday is 170, then find the number of salad bowls sold on Friday.

यदि सोमवार, मंगलवार, बुधवार, गुरुवार और शुक्रवार को बेचे गए सलाद बाउल की औसत संख्या 170 है, तो शुक्रवार को बेचे गए सलाद बाउल की संख्या ज्ञात कीजिए।

- (A) 220
 (B) 210
 (C) 230
 (D) 240
 (E) 200

Q5: On Sunday, the number of sandwiches and juice bottles sold increased by 20% and 10% respectively as compared to Tuesday, while the total number of food items sold on Sunday remained the same as Tuesday. Find the percentage decrease in the sale of salad bowls on Sunday as compared to Tuesday.

रविवार को बेचे गए सैंडविच और जूस बोतलों की संख्या मंगलवार की तुलना में क्रमशः 20% और 10% बढ़ गई, जबकि रविवार को बेचे गए कुल खाद्य पदार्थों की संख्या मंगलवार के समान रही। मंगलवार की तुलना में रविवार को सलाद बाउल की बिक्री में प्रतिशत कमी ज्ञात कीजिए।

- (A) 45%
 (B) 35%
 (C) 30%
 (D) 50%
 (E) 40%

Solutions:

Particulars	Monday	Tuesday	Wednesday	Thursday
Average of 3 items	200	225	175	250
Sandwiches sold	250	225	180	260
Difference (Juice - Salad)	110	90	85	90
Total food items	600	675	525	750
Juice bottles sold	230	270	215	290
Salad bowls sold	120	180	130	200

1. Option D

Total food items sold on Monday = $200 \times 3 = 600$

Sandwiches sold on Monday = 250

$$\text{Juice bottles} + \text{Salad bowls} = 600 - 250 = 350$$

$$\text{Juice bottles} - \text{Salad bowls} = 110$$

$$\text{Salad bowls sold on Monday} = (350 - 110) \div 2 = 120$$

$$\text{Total food items sold on Saturday} = 600 \times 80\% = 480$$

$$\text{Salad bowls sold on Saturday} = 120$$

$$\text{Sandwiches} + \text{Juice bottles on Saturday} = 480 - 120 = 360$$

$$\text{Sandwiches} : \text{Juice bottles} = 7 : 5$$

$$\text{Sandwiches sold on Saturday} = 360 \times 7 \div 12 = 210$$

2. Option B

$$\text{Total food items sold on Wednesday} = 175 \times 3 = 525$$

$$\text{Sandwiches sold on Wednesday} = 180$$

$$\text{Juice bottles} + \text{Salad bowls} = 525 - 180 = 345$$

$$\text{Juice bottles} - \text{Salad bowls} = 85$$

$$\text{Juice bottles sold on Wednesday} = (345 + 85) \div 2 = 215$$

$$\text{Salad bowls sold on Wednesday} = (345 - 85) \div 2 = 130$$

$$\text{Total revenue} = 180 \times 12 + 215 \times 8 + 130 \times A$$

$$5700 = 2160 + 1720 + 130A$$

$$130A = 1820$$

$$A = 14$$

3. Option C

$$\text{Total food items sold on Tuesday} = 225 \times 3 = 675$$

$$\text{Sandwiches sold on Tuesday} = 225$$

$$\text{Juice bottles} + \text{Salad bowls} = 675 - 225 = 450$$

$$\text{Juice bottles} - \text{Salad bowls} = 90$$

$$\text{Juice bottles sold on Tuesday} = (450 + 90) \div 2 = 270$$

$$\text{Salad bowls sold on Tuesday} = (450 - 90) \div 2 = 180$$

$$\text{Required percentage} = 270 \times 100 \div (225 + 180)$$

$$\text{Required percentage} = 270 \times 100 \div 405 = 66 \frac{2}{3}\%$$

4. Option A

$$\text{Salad bowls on Monday} = 120$$

$$\text{Salad bowls on Tuesday} = 180$$

$$\text{Salad bowls on Wednesday} = 130$$

$$\text{Total food items sold on Thursday} = 250 \times 3 = 750$$

$$\text{Sandwiches sold on Thursday} = 260$$

$$\text{Juice bottles} + \text{Salad bowls} = 750 - 260 = 490$$

Juice bottles - Salad bowls = 90

Salad bowls on Thursday = $(490 - 90) \div 2 = 200$

Total salad bowls in five days = $170 \times 5 = 850$

Salad bowls on Friday = $850 - (120 + 180 + 130 + 200)$

Salad bowls on Friday = 220

5. Option E

Total food items sold on Tuesday = 675

Sandwiches sold on Tuesday = 225

Juice bottles sold on Tuesday = 270

Salad bowls sold on Tuesday = 180

Sandwiches sold on Sunday = $225 \times 120\% = 270$

Juice bottles sold on Sunday = $270 \times 110\% = 297$

Salad bowls sold on Sunday = $675 - 270 - 297 = 108$

Decrease in salad bowls = $180 - 108 = 72$

Required percentage decrease = $72 \times 100 \div 180 = 40\%$

Set 2: Five spice marts sold three types of spice packets: turmeric, chilli and coriander only. The table shows the ratio of turmeric, chilli and coriander packets sold respectively by each spice mart. It also shows the average number of turmeric and chilli packets sold by each mart. Read the table carefully and answer the questions.

पाँच मसाला मार्टों ने केवल तीन प्रकार के मसाला पैकेट बेचे: हल्दी, मिर्च और धनिया। तालिका प्रत्येक मसाला मार्ट द्वारा क्रमशः बेचे गए हल्दी, मिर्च और धनिया पैकेटों का अनुपात दर्शाती है। यह प्रत्येक मार्ट द्वारा बेचे गए हल्दी और मिर्च पैकेटों की औसत संख्या भी दर्शाती है। तालिका को ध्यानपूर्वक पढ़िए और प्रश्नों के उत्तर दीजिए।

Spice Mart / मसाला मार्ट	Ratio of turmeric, chilli and coriander packets sold respectively / बेचे गए हल्दी, मिर्च और धनिया पैकेटों का क्रमशः अनुपात	Average number of turmeric and chilli packets sold / बेचे गए हल्दी और मिर्च पैकेटों की औसत संख्या
Varanasi Mart / वाराणसी मार्ट	4:5:6	162
Madurai Mart / मदुरै मार्ट	5:7:4	216
Kozhikode Mart / कोझिकोड मार्ट	7:3:5	180
Siliguri Mart / सिलीगुड़ी मार्ट	2:5:7	126
Udaipur Mart / उदयपुर मार्ट	7:4:5	198

Q1: Find the average number of turmeric packets sold by all the given spice marts.
सभी दिए गए मसाला मार्टों द्वारा बेचे गए हल्दी पैकेटों की औसत संख्या ज्ञात कीजिए।

- (A) 174
(B) 186
(C) 180
(D) 192
(E) 168

Q2: The average weight of one turmeric packet, one chilli packet and one coriander packet is 150 gm, 100 gm and 200 gm respectively. Find the total weight, in kg, of all the packets sold by Madurai Mart.

एक हल्दी पैकेट, एक मिर्च पैकेट और एक धनिया पैकेट का औसत वजन क्रमशः 150 ग्राम, 100 ग्राम और 200 ग्राम है। मदुरै मार्ट द्वारा बेचे गए सभी पैकेटों का कुल वजन, किलोग्राम में, ज्ञात कीजिए।

- (A) 77
(B) 84
(C) 79

- (D) 87
(E) 81

Q3: If the ratio of chilli packets sold by Kozhikode Mart and Udaipur Mart together to coriander packets sold by Varanasi Mart and Kozhikode Mart together is $P:Q$, then find $P + Q$.

यदि कोझिकोड मार्ट और उदयपुर मार्ट द्वारा मिलाकर बेचे गए मिर्च पैकेटों तथा वाराणसी मार्ट और कोझिकोड मार्ट द्वारा मिलाकर बेचे गए धनिया पैकेटों का अनुपात $P:Q$ है, तो $P + Q$ ज्ञात कीजिए।

- (A) 20
(B) 18
(C) 24
(D) 22
(E) 16

Q4: The number of turmeric packets sold by Solapur Mart is 25% less than the total number of chilli and coriander packets sold by Siliguri Mart. If the ratio of turmeric, chilli and coriander packets sold by Solapur Mart is $3:4:5$ respectively, then find the number of coriander packets sold by Solapur Mart.

सोलापुर मार्ट द्वारा बेचे गए हल्दी पैकेटों की संख्या, सिलीगुड़ी मार्ट द्वारा बेचे गए मिर्च और धनिया पैकेटों की कुल संख्या से 25% कम है। यदि सोलापुर मार्ट द्वारा बेचे गए हल्दी, मिर्च और धनिया पैकेटों का अनुपात क्रमशः $3:4:5$ है, तो सोलापुर मार्ट द्वारा बेचे गए धनिया पैकेटों की संख्या ज्ञात कीजिए।

- (A) 504
(B) 576
(C) 522
(D) 540
(E) 558

Q5: The number of chilli and coriander packets together sold by Varanasi Mart is what percentage of the number of coriander packets sold by Udaipur Mart?

वाराणसी मार्ट द्वारा मिलाकर बेचे गए मिर्च और धनिया पैकेट, उदयपुर मार्ट द्वारा बेचे गए धनिया पैकेटों का कितने प्रतिशत हैं?

- (A) 220%
(B) 240%
(C) 200%
(D) 180%

(E) 260%

Solutions:

Spice Mart / मसाला मार्ट	Turmeric / हल्दी	Chilli / मिर्च	Coriander / धनिया
Varanasi / वाराणसी	144	180	216
Madurai / मदुरै	180	252	144
Kozhikode / कोझिकोड	252	108	180
Siliguri / सिलीगुड़ी	72	180	252
Udaipur / उदयपुर	252	144	180

Common values:

Varanasi Mart: Turmeric + Chilli = $162 \times 2 = 324$ $4 + 5 = 9$ parts = 324, so 1 part = 36

Turmeric = 144, Chilli = 180, Coriander = 216

Madurai Mart: Turmeric + Chilli = $216 \times 2 = 432$ $5 + 7 = 12$ parts = 432, so 1 part = 36

Turmeric = 180, Chilli = 252, Coriander = 144

Kozhikode Mart: Turmeric + Chilli = $180 \times 2 = 360$ $7 + 3 = 10$ parts = 360, so 1 part = 36

Turmeric = 252, Chilli = 108, Coriander = 180

Siliguri Mart: Turmeric + Chilli = $126 \times 2 = 252$ $2 + 5 = 7$ parts = 252, so 1 part = 36

Turmeric = 72, Chilli = 180, Coriander = 252

Udaipur Mart: Turmeric + Chilli = $198 \times 2 = 396$ $7 + 4 = 11$ parts = 396, so 1 part = 36

Turmeric = 252, Chilli = 144, Coriander = 180

1. Option CTurmeric packets sold by all marts = $144 + 180 + 252 + 72 + 252 = 900$ Average = $900/5 = 180$ **2. Option E**

Madurai Mart: Turmeric = 180, Chilli = 252, Coriander = 144

Total weight = $180 \times 150 + 252 \times 100 + 144 \times 200$ Total weight = $27000 + 25200 + 28800 = 81000$ gm

Total weight = 81 kg

3. Option B

Chilli packets in Kozhikode and Udaipur = $108 + 144 = 252$

Coriander packets in Varanasi and Kozhikode = $216 + 180 = 396$

Ratio = $252:396 = 7:11$

$P + Q = 7 + 11 = 18$

4. Option D

Chilli and coriander packets in Siliguri = $180 + 252 = 432$

Turmeric packets in Solapur = 75% of $432 = 324$

3 parts = 324, so 1 part = 108

Coriander packets in Solapur = $5 \times 108 = 540$

5. Option A

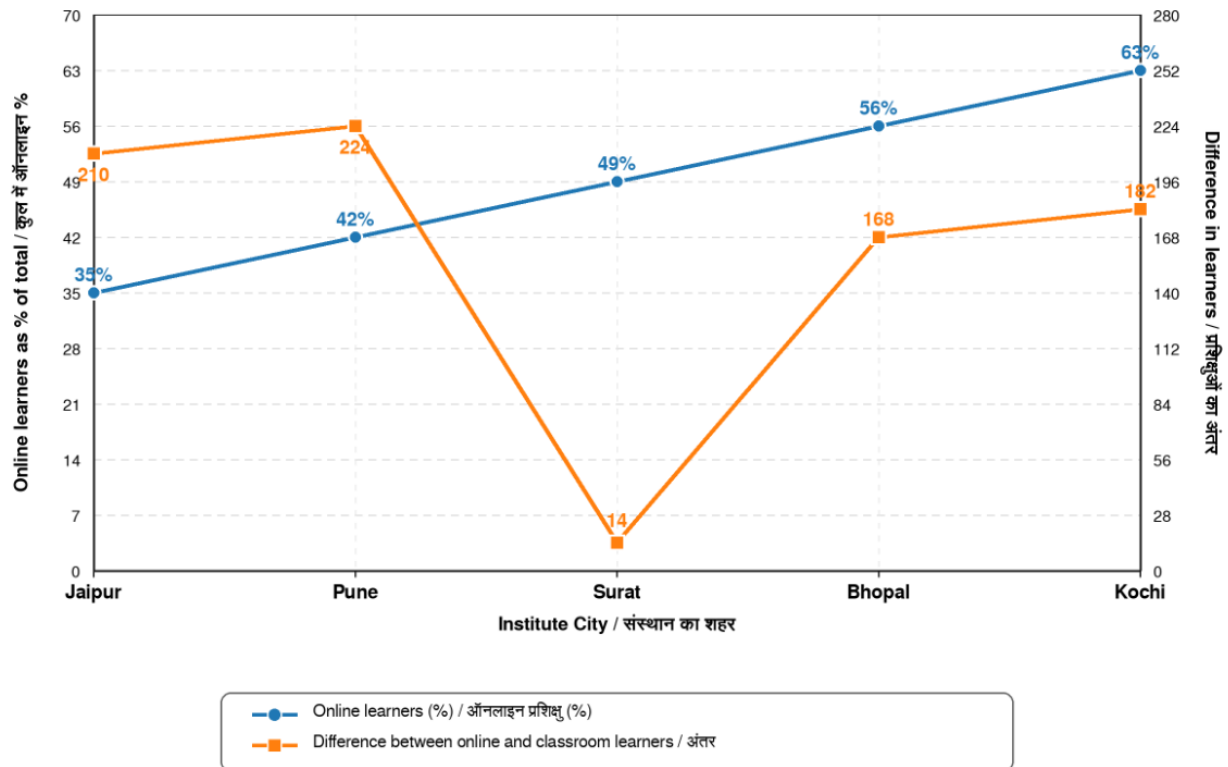
Chilli and coriander packets in Varanasi = $180 + 216 = 396$

Coriander packets in Udaipur = 180

Required percentage = $396/180 \times 100 = 220\%$

Set 3: Five skill-training institutes in Jaipur, Pune, Surat, Bhopal and Kochi have different numbers of learners in two modes: online and classroom. The line chart shows the number of online learners as a percentage of the total number of learners in each institute. It also shows the positive difference between the number of online and classroom learners in each institute. Study the data carefully and answer the questions.

जयपुर, पुणे, सूरत, भोपाल और कोच्चि के पांच कौशल-प्रशिक्षण संस्थानों में ऑनलाइन और कक्षा मोड में अलग-अलग संख्या में प्रशिक्षु हैं। रेखा आलेख प्रत्येक संस्थान में कुल प्रशिक्षुओं में ऑनलाइन प्रशिक्षुओं का प्रतिशत दर्शाता है। यह प्रत्येक संस्थान में ऑनलाइन और कक्षा प्रशिक्षुओं की संख्या के बीच का धनात्मक अंतर भी दर्शाता है। डेटा का ध्यानपूर्वक अध्ययन करें और प्रश्नों के उत्तर दें।



Q1: The ratio of the number of online learners in Pune and Surat together to the number of classroom learners in the same two cities together is $x:y$. Find $x + y$.

पुणे और सूरत में मिलाकर ऑनलाइन प्रशिक्षुओं की संख्या तथा उन्हीं दोनों शहरों में मिलाकर कक्षा प्रशिक्षुओं की संख्या का अनुपात $x:y$ है। $x + y$ ज्ञात कीजिए।

- (A) 286
- (B) 312
- (C) 300
- (D) 324
- (E) 294

Q2: Each online learner and each classroom learner receives an average of 42 and 63 practice sheets respectively. Find the total number of practice sheets required for all learners in Pune.

प्रत्येक ऑनलाइन प्रशिक्षु और प्रत्येक कक्षा प्रशिक्षु को क्रमशः औसतन 42 और 63 अभ्यास पत्र मिलते हैं। पुणे के सभी प्रशिक्षुओं के लिए आवश्यक कुल अभ्यास पत्रों की संख्या ज्ञात कीजिए।

- (A) 74256
- (B) 76734
- (C) 73962
- (D) 77616
- (E) 75852

Q3: The number of online learners in Jaipur and Bhopal together is what percent of the total number of learners in Jaipur?

जयपुर और भोपाल में मिलाकर ऑनलाइन प्रशिक्षुओं की संख्या, जयपुर के कुल प्रशिक्षुओं की संख्या का कितना प्रतिशत है?

- (A) 147%
- (B) 143%
- (C) 151%
- (D) 154%
- (E) 139%

Q4: The average number of classroom learners in Bhopal, Kochi and Indore is 525. If $58\frac{1}{3}\%$ of the learners in Indore are online learners, then find the number of online learners in Indore.

भोपाल, कोच्चि और इंदौर में कक्षा प्रशिक्षुओं की औसत संख्या 525 है। यदि इंदौर के $58\frac{1}{3}\%$ प्रशिक्षु ऑनलाइन प्रशिक्षु हैं, तो इंदौर में ऑनलाइन प्रशिक्षुओं की संख्या ज्ञात कीजिए।

- (A) 840
- (B) 910
- (C) 1050
- (D) 980
- (E) 1120

Q5: The difference between the sum of classroom learners in Jaipur and Pune together and the sum of online learners in Pune and Surat together is k. Find $112\frac{1}{2}\%$ of k.

जयपुर और पुणे में मिलाकर कक्षा प्रशिक्षुओं की संख्या के योग तथा पुणे और सूरत में

मिलाकर ऑनलाइन प्रशिक्षुओं की संख्या के योग का अंतर k है। k का $112\frac{1}{2}\%$ ज्ञात कीजिए।

- (A) 364
- (B) 378
- (C) 406
- (D) 434
- (E) 462

Solutions:

1. Option C

$$\text{Pune total} = 224 \times 100/16 = 1400$$

$$\text{Pune online} = 42\% \text{ of } 1400 = 588 \text{ and classroom} = 812$$

$$\text{Surat total} = 14 \times 100/2 = 700$$

$$\text{Surat online} = 49\% \text{ of } 700 = 343 \text{ and classroom} = 357$$

$$\text{Required ratio} = (588 + 343):(812 + 357) = 931:1169 = 133:167$$

$$x + y = 133 + 167 = 300$$

2. Option E

$$\text{Pune total} = 224 \times 100/16 = 1400$$

$$\text{Pune online} = 42\% \text{ of } 1400 = 588 \text{ and classroom} = 812$$

$$\text{Required practice sheets} = 588 \times 42 + 812 \times 63$$

$$\text{Required practice sheets} = 24696 + 51156 = 75852$$

3. Option A

$$\text{Jaipur total} = 210 \times 100/30 = 700$$

$$\text{Jaipur online} = 35\% \text{ of } 700 = 245$$

$$\text{Bhopal total} = 168 \times 100/12 = 1400$$

$$\text{Bhopal online} = 56\% \text{ of } 1400 = 784$$

$$\text{Required percentage} = (245 + 784)/700 \times 100$$

$$\text{Required percentage} = 1029/700 \times 100 = 147\%$$

4. Option D

$$\text{Bhopal classroom learners} = 44\% \text{ of } 1400 = 616$$

$$\text{Kochi total} = 182 \times 100/26 = 700$$

$$\text{Kochi classroom learners} = 37\% \text{ of } 700 = 259$$

$$\text{Total classroom learners in Bhopal, Kochi and Indore} = 525 \times 3 = 1575$$

$$\text{Indore classroom learners} = 1575 - 616 - 259 = 700$$

In Indore, online learners = $58\frac{1}{3}\%$ and classroom learners = $41\frac{2}{3}\%$

Online:Classroom = 7:5

Indore online learners = $700 \times \frac{7}{5} = 980$

5. Option B

Jaipur classroom learners = 65% of 700 = 455

Pune classroom learners = 812

Pune online learners = 588

Surat online learners = 343

$k = (455 + 812) - (588 + 343)$

$k = 1267 - 931 = 336$

$112\frac{1}{2}\%$ of $k = 336 \times \frac{9}{8} = 378$

Set 4: The data is about the number of handmade greeting cards sold by three craft stalls, Nidhi Stall, Gaurav Stall and Kavya Stall, in two different years, 2022 and 2023. The ratio of greeting cards sold by Nidhi Stall in 2022 to greeting cards sold by Kavya Stall in 2023 is 3 : 2. The number of greeting cards sold by Nidhi Stall in 2023 is 25% more than the number of greeting cards sold by Kavya Stall in 2023. The number of greeting cards sold by Gaurav Stall in 2022 is equal to the number of greeting cards sold by Nidhi Stall in 2023. The number of greeting cards sold by Gaurav Stall in 2023 is 25% more than the number of greeting cards sold by Kavya Stall in 2022. The total number of greeting cards sold by all three stalls in 2022 is 448, and the average number of greeting cards sold by all three stalls in 2023 is 136.

यह डेटा तीन क्राफ्ट स्टॉल, निधि स्टॉल, गौरव स्टॉल और काव्या स्टॉल, द्वारा दो अलग-अलग वर्षों 2022 और 2023 में बेचे गए हस्तनिर्मित ग्रीटिंग कार्ड्स की संख्या से संबंधित है। 2022 में निधि स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स और 2023 में काव्या स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स का अनुपात 3 : 2 है। 2023 में निधि स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स की संख्या, 2023 में काव्या स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स की संख्या से 25% अधिक है। 2022 में गौरव स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स की संख्या, 2023 में निधि स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स की संख्या के बराबर है। 2023 में गौरव स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स की संख्या, 2022 में काव्या स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स की संख्या से 25% अधिक है। 2022 में तीनों स्टॉल द्वारा बेचे गए कुल ग्रीटिंग कार्ड्स की संख्या 448 है, और 2023 में तीनों स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स की औसत संख्या 136 है।

Q1: Greeting cards sold by Kavya Stall in 2023 are p% of greeting cards sold by Gaurav Stall in 2022. Find the value of p.

2023 में काव्या स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स, 2022 में गौरव स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स का p% हैं। p का मान ज्ञात कीजिए।

- (A) 72
- (B) 76
- (C) 80
- (D) 84
- (E) 88

Q2: If greeting cards sold by Manas Stall in 2023 are 25% less than the greeting cards sold by Nidhi Stall in 2022, then find the total number of greeting cards sold by Nidhi Stall, Gaurav Stall, Kavya Stall and Manas Stall together in 2023.

यदि 2023 में मानस स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स, 2022 में निधि स्टॉल द्वारा बेचे गए

ग्रीटिंग कार्ड्स से 25% कम हैं, तो 2023 में निधि स्टॉल, गौरव स्टॉल, काव्या स्टॉल और मानस स्टॉल द्वारा मिलाकर बेचे गए कुल ग्रीटिंग कार्ड्स की संख्या ज्ञात कीजिए।

- (A) 516
- (B) 528
- (C) 540
- (D) 564
- (E) 552

Q3: Find the difference between the average number of greeting cards sold by Kavya Stall and Gaurav Stall in both the years.

दोनों वर्षों में काव्या स्टॉल और गौरव स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स की औसत संख्या के बीच अंतर ज्ञात कीजिए।

- (A) 24
- (B) 28
- (C) 32
- (D) 36
- (E) 40

Q4: If the ratio of greeting cards sold by Nidhi Stall in 2024 to greeting cards sold by Nidhi Stall in 2023 is 4 : 5, then greeting cards sold by Kavya Stall in 2022 are p% less than greeting cards sold by Nidhi Stall in 2024. Find p.

यदि 2024 में निधि स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स और 2023 में निधि स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स का अनुपात 4 : 5 है, तो 2022 में काव्या स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स, 2024 में निधि स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स से p% कम हैं। p का मान ज्ञात कीजिए।

- (A) 15
- (B) 20
- (C) 30
- (D) 25
- (E) 35

Q5: In 2023, 25% of the greeting cards sold by Gaurav Stall and 12.5% of the greeting cards sold by Kavya Stall had printing defects. Find the total number of non-defective greeting cards sold by Gaurav Stall and Kavya Stall together in 2023.

2023 में गौरव स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स में से 25% और काव्या स्टॉल द्वारा बेचे गए ग्रीटिंग कार्ड्स में से 12.5% में प्रिंटिंग की खराबी थी। 2023 में गौरव स्टॉल और काव्या स्टॉल द्वारा मिलाकर बेचे गए बिना खराबी वाले ग्रीटिंग कार्ड्स की कुल संख्या ज्ञात कीजिए।

- (A) 202

- (B) 198
 (C) 214
 (D) 208
 (E) 192

Solutions:

Stall	2022	2023	Total
Nidhi Stall	192	160	352
Gaurav Stall	160	120	280
Kavya Stall	96	128	224
Total	448	408	856

1. Option C

Greeting cards sold by Kavya Stall in 2023 = 128

Greeting cards sold by Gaurav Stall in 2022 = 160

$$p = 128 \div 160 \times 100$$

$$= 80$$

2. Option E

Greeting cards sold by Nidhi Stall in 2022 = 192

Greeting cards sold by Manas Stall in 2023 = 75% of 192

$$= 144$$

Total cards sold by three given stalls in 2023 = 408

Required total = 408 + 144

$$= 552$$

3. Option B

Average cards sold by Kavya Stall = $(96 + 128) \div 2$

$$= 112$$

Average cards sold by Gaurav Stall = $(160 + 120) \div 2$

$$= 140$$

Required difference = 140 – 112

$$= 28$$

4. Option D

Greeting cards sold by Nidhi Stall in 2023 = 160

Nidhi Stall in 2024 : Nidhi Stall in 2023 = 4 : 5

Greeting cards sold by Nidhi Stall in 2024 = $160 \times 4 \div 5$
= 128

Greeting cards sold by Kavya Stall in 2022 = 96

$p = (128 - 96) \div 128 \times 100$
= 25

5. Option A

Greeting cards sold by Gaurav Stall in 2023 = 120

Non-defective cards of Gaurav Stall = 75% of 120
= 90

Greeting cards sold by Kavya Stall in 2023 = 128

Non-defective cards of Kavya Stall = 87.5% of 128
= 112

Required total = 90 + 112
= 202